

V3KTOR — Mapping the Self

An affective calibration instrument: method, design rationale, and theoretical foundation

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Abstract

V3KTOR is an affective calibration instrument that measures a user's instinctive emotional reactions to a curated set of words by recording binary responses (positive / negative) and response time in milliseconds. The instrument applies the Valence-Arousal-Dominance (VAD) model from affective science (Mehrabian & Russell, 1974; Russell, 1980), grounded in published affective norms (Bradley & Lang, 1999, ANEW; Mohammad, 2018, NRC VAD Lexicon; Monnier & Syssau, 2014, FAN), and treats response time as a signal of automaticity following the lexical-decision and Implicit Association Test (IAT) literature (Greenwald, McGhee & Schwartz, 1998; Balota et al., 2007).

V3KTOR is explicitly framed as an *instrument*, not a therapeutic or diagnostic tool. Its function is to produce a reading on demand and remain inert otherwise. The output is a structured session record that the user controls and exports to external large-language-model (LLM) systems for interpretation. The instrument's novelty lies not in the underlying VAD framework, which is public science, but in the combination of (a) reflex-time as the primary signal axis applied to personal reflection rather than research, (b) a set of session-integrity safeguards designed to preserve signal quality, and (c) a portable, user-owned session format designed for downstream interpretation by both human readers and AI agents.

This document describes the method, the specific design decisions, and the rationale for each, as of the date above. It is deposited as prior art and as the authoritative record of the V3KTOR method on this date.

1. Purpose and Philosophy

V3KTOR is built on a single design premise: the instrument's job is to give an honest reading when asked, and then stop. It is not a companion, a coach, or a wellness intervention. It does not push notifications, suggest exercises, or attempt to retain the user beyond the moment a reading is needed. The analogy used throughout the design process is the blood pressure cuff: accurate on demand, invisible otherwise. The instrument earns return visits through fidelity, not through engagement mechanics.

This framing is non-negotiable in the design and has shaped every subsequent decision. Features that would be standard in adjacent product categories — daily streaks, gamification, AI-companion dialogue, in-app advice, personalized content feeds — are deliberately excluded. The free version of the instrument is the complete instrument; no premium tier exists to gate the reading itself.

The thesis underlying the instrument is that contemporary humans receive most signal in a day about other people — algorithmic feeds, social content, commercial messaging, agendas of others. This is a recent environmental condition; the human nervous system evolved for an environment where most signal came from immediate surroundings and concerned the individual's own life. In the current environment, the default state is drift: the user absorbs whichever narrative is most prevalent and, over time, may begin to mistake it for their own voice. A baseline instrument — a measured, repeatable

reading of stable affective response — produces a reference point against which drift becomes detectable.

2. Theoretical Foundation

2.1 The VAD model

The Valence-Arousal-Dominance (VAD) model originates with Mehrabian & Russell (1974) and Russell's circumplex model (1980). It is among the most established dimensional frameworks for describing affective states. V3KTOR uses VAD as the descriptive coordinate system for both individual word stimuli and aggregated session output:

Valence (V): the pleasantness/unpleasantness axis. Values below 0.35 are treated as negative-band, above 0.65 as positive-band, between as neutral. **Arousal (A):** the activation axis (calm to excited). **Dominance (D):** the control axis (submissive to dominant). Each word in the V3KTOR lexicon carries pre-assigned V, A, D values drawn from published affective norms.

2.2 Word value sources

Word values align with the NRC VAD Lexicon (Mohammad, 2018), the Affective Norms for English Words (ANEW; Bradley & Lang, 1999), and the French Affective Norms (FAN; Monnier & Syssau, 2014) for the French-language version. These are standard references in affective sciences and are used to assign V, A, D scores to each stimulus word.

2.3 Response time as automaticity signal

The use of response time as a signal of emotional automaticity draws on lexical-decision research (Balota et al., 2007) and the Implicit Association Test (IAT) paradigm (Greenwald, McGhee & Schwartz, 1998). This literature establishes that responses under approximately 500–700ms typically reflect automatic rather than deliberate processing. V3KTOR uses this finding operationally: faster responses are interpreted as reflexive, slower responses as friction or deliberation.

Concretely, the instrument calibrates thresholds against the user's own response window (see §3.1). The default response window is 2000ms; thresholds are then derived as proportions of that window: responses under 30% of max-RT are flagged as automatic; responses above 60% as friction; intermediate as deliberate.

2.4 Stability premise

The instrument assumes that adult affective response, especially in the value domain, is sufficiently stable to be meaningfully mapped over repeated readings. This is supported by decades of longitudinal personality research showing rank-order stability of .60–.80 across multi-year intervals (McCrae & Costa, 1994; Roberts & DelVecchio, 2000; Borghuis et al., 2017) and by Schwartz value-hierarchy stability across major life transitions (Vecchione et al., 2016). The instrument does not claim affective response is fixed — meaningful change continues across the lifespan (Roberts, Walton & Viechtbauer, 2006) — only that the underlying structure moves slowly enough that session-to-session fluctuation is mostly noise against a stable map, rather than the map itself redrawing.

3. Instrument Design

3.1 Personal response-time calibration

Default thresholds (under 30% of max-RT = automatic; over 60% = friction) are calibrated against a population average. V3KTOR additionally maintains a per-user median response time stored locally

(key: tc_personal_rt_v1) and blends this personal median into the threshold calculation using an exponentially weighted moving average (EWMA) with a 70/30 ratio (70% prior personal median, 30% current session median). The EWMA blend begins after 5 valid sessions and stabilises by 10 sessions.

Rationale: motor and cognitive response times vary substantially between users due to age, device input modality, hand-eye coordination, baseline arousal, and momentary distractions. A fixed-threshold model misclassifies fast-but-deliberate responders as reflexive and slow-but-reflexive responders as conflicted. Personal calibration corrects this by adapting to the user's own response distribution.

3.2 Reflex-drift detector

After a minimum of 5 saved sessions, the instrument computes the median average-RT across the user's 5 most recent sessions. If this median drifts substantially below or above the currently calibrated thresholds, the instrument surfaces an optional adjustment suggestion ("REFLEX SHARPENED" or "REFLEX SLOWED") proposing a new response window. A 10-session cooldown is applied after each surfacing to prevent re-triggering on close cases.

Trigger thresholds: 15% below the fast threshold for sharpening, 10% above the slow threshold for slowing. The suggested new window is computed as $2 \times$ the user's current median, rounded to the nearest 100ms, clamped to the settings slider range [1200, 2500]ms.

Rationale: the user's reflex baseline can shift over weeks or months due to fatigue, practice effects, neurological changes, medication, age, or environmental shifts. Without drift detection, the instrument's classifications gradually drift out of correspondence with the user's actual response distribution, degrading signal quality. The drift detector surfaces this when it crosses a meaningful threshold, while remaining inert when the user's baseline is stable.

3.3 Band-balancing constraint

The word selection algorithm enforces a hard constraint that no more than 3 consecutive same-valence-band words appear in a session (negative band: $V < 0.35$; positive band: $V > 0.65$). The instrument tracks shown bands and blocks any candidate that would create a 4-in-a-row of either band. After 4 consecutive positive responses, the instrument also injects a pivot word followed by a dark-cluster word.

Rationale: extended runs of same-valence stimuli produce two distinct contamination effects. First, motor-memory adaptation: the user begins tapping the same side reflexively, which destroys the response-time signal for that band. Second, mood induction: emotionally coherent runs of words can shift the user's transient state, contaminating subsequent responses. Band-balancing preserves the discriminative power of each individual response.

3.4 Stratified interleaving (STRIDE)

The initial word pool is constructed by stratified sampling across 12 thematic clusters (relational, grief, alone, transcendence, dark, light, power-dark, power-light, primal, nature, transform, pivot). Every third word in the session is drawn directly from this stratified pool rather than from the adaptive queue, ensuring pool integrity is preserved even when the adaptive selection would otherwise dominate.

Rationale: a purely adaptive word selector tends to over-sample bands the user responds strongly to, producing a skewed signal. Stratified interleaving guarantees that every session probes all clusters with sufficient density to detect cross-cluster patterns.

3.5 Pivot words and light-side failsafe

Pivot words are a designated cluster of 'turning-point' stimuli (words like 'change', 'release', 'forgive', 'begin') inserted after a sustained negative or positive run. The user's response to the pivot word —

accepted (positive) versus held at distance (negative) — is recorded separately as a signal of the user's relationship to change in that affective context.

The light-side failsafe is the positive-band mirror of the existing dark-streak mechanic. After 4 consecutive positive responses, the instrument injects a pivot word followed by a dark-cluster word, preventing the session from becoming an unbroken affirmation loop. Rationale: unbroken positive runs are as motor-contaminated and as mood-inductive as unbroken negative runs, and produce equally degraded signal.

3.6 Conflict detection

When the user's response inverts a word's pre-assigned valence (responding positively to a negative-band word, or negatively to a positive-band word), the response is recorded as a 'conflict.' Conflicts are surfaced separately in the session output and the AI handoff prefix as a key interpretive signal: the user's instinctive reaction has diverged from the word's normative affective charge.

3.7 Mood declaration (pre-test)

Before each test, the user optionally selects one of six declared moods (calm, happy, sad, anxious, angry, tired). The declared mood is stored on the session object and included in the export. Its purpose is to allow comparison between the user's *story* about their current state and the *reflex* measured by the instrument; persistent mood-vs-reflex divergence is itself a signal.

4. AI Handoff Design

V3KTOR is not an interpretation engine. The instrument produces a structured session record; interpretation is delegated to the user, optionally assisted by external LLM systems (Claude, ChatGPT, Gemini, etc.) via a Copy-for-AI export. The export bundles the session data with a carefully designed prompt prefix (the 'AI handoff') that frames the interpretation task for the receiving model. The handoff design encodes several explicit safety and quality constraints, described below.

4.1 Portrait-first opening

The handoff instructs the receiving model to begin its response with a qualitative character portrait of the user inferred from the data, rather than restating the metrics. Rationale: a metric-restatement opening reads as performative summarization and fails to produce interpretive value. A portrait-first opening forces the model to synthesize before summarizing, which produces a substantively useful first paragraph and signals to the user that the model has actually engaged with the data.

4.2 Cross-section vs longitudinal framing

The handoff distinguishes single-session readings (cross-sectional) from aggregated baselines (longitudinal). For single sessions, the receiving model is instructed to treat the data as a snapshot of present state, vulnerable to noise. For baselines, the model is instructed to treat the data as a portrait of stable structure, with deviations between individual sessions interpreted as noise against the stable map.

4.3 Recursion governor

The handoff includes an explicit constraint that subsequent passes (where the user asks follow-up questions) must re-anchor in previously unused data fields, rather than recursing deeper into the same interpretive thread. The model is directed not to invite the user with open-ended 'want to go deeper?' prompts unless the user explicitly requests more.

Rationale: empirical observation during V3KTOR's development showed that some LLMs (notably DeepSeek in testing) produce escalating recursive prompts, repeatedly inviting the user to elaborate on the most emotionally charged thread. For users in distress, this pattern can lead deeper into difficult affective territory rather than producing useful reflection. The recursion governor is an explicit countermeasure: it instructs the model to broaden, not deepen, on subsequent turns.

4.4 Distress-signal close

The handoff includes a closing instruction for the model to surface appropriate support resources if specific risk patterns appear in the session data (e.g., extremely fast responses to high-arousal negative stimuli combined with conflict patterns on light-cluster words). The close is keyed to data patterns, not to the user's declared mood, so the model does not flag distress purely on self-report.

Rationale: as an instrument deployed without clinical supervision and read by LLMs of varying alignment quality, V3KTOR carries some duty of care that crisis signals not be ignored. The distress-signal close is the minimal viable safeguard: it does not interrupt normal readings but ensures that genuinely concerning patterns produce an appropriate closing note from the interpreting model.

5. Session Export Format (Summary)

The session export is a plain-text, human-readable record produced by the Copy-for-AI button. It is designed for portability — to be pasted into any LLM chat interface — and for direct human inspection. The full specification is described in the companion document *V3KTOR Session File Format Specification v1.0* (Tobie David, 2026), deposited separately. The export includes: ISO date and time; declared mood; aggregated VAD percentages; calibrated response-time thresholds; word-level data (positive set, negative set, conflicts, fastest responses) with response times; pivot word and outcome where present.

The export is designed to be future-extended into a signed, portable affective profile (see §6). At the present version (v1), the export is plain text with a documented schema and is not yet cryptographically signed.

6. Phase 2 Vision: The Affective Alignment File

The longer-term ambition of V3KTOR is to evolve the session export into a portable, cryptographically signed, user-owned *affective profile* intended for use as a personalization input to local AI agents. The premise is that current AI personalization is built on behavioural data (clicks, purchases) and preference data (surveys, ratings), both of which are gameable and shallow. Reflex response time to affective stimuli is qualitatively different: it is difficult to fake, low-dimensional, and stable enough to be reused across sessions and devices.

A V3KTOR alignment file would contain, in compact form: the user's stable VAD signature across longitudinal sessions, their characteristic conflict patterns, their reflex thresholds, and a cryptographic signature anchoring authorship. The file format would be designed to be read by local AI agents (e.g., personal LLM assistants running on-device) to inform tone, framing, content selection, and trust calibration — without the agent needing access to the user's full behavioural history or message logs.

This vision is recorded here as part of the V3KTOR roadmap as of the date of this document. The specific signing scheme and binary format will be specified separately as the implementation matures.

7. Distinguishing Properties (Summary)

The V3KTOR method is distinguished from prior affective-science instruments and from commercial mood-tracking applications by the combination of the following properties, all of which are present in the May 2026 reference implementation:

(a) Reflex response time as the primary signal axis, applied to personal reflection rather than research; (b) per-user EWMA-blended response-time calibration with a documented 5–10 session ramp; (c) the reflex-drift detector with cooldown logic; (d) the hard band-balancing constraint and the symmetric light-side failsafe; (e) stratified interleaving (STRIDE) to preserve pool integrity against adaptive selection drift; (f) the pivot-word mechanic for measuring relationship to change; (g) the portrait-first, cross-section/longitudinal-distinguishing AI handoff prefix; (h) the recursion governor constraint on follow-up interpretation; (i) the data-keyed distress-signal close; (j) the portable plain-text session export designed for both human and AI consumption; (k) the explicit instrument-not-companion design philosophy and its concrete absence-of-engagement-mechanics expression in the product.

8. Statement on Scientific Status

V3KTOR is not a clinical or diagnostic instrument. It is a personal reflection tool that applies a recognized scientific framework (VAD; reaction-time as automaticity signal) to personal use. The instrument does not diagnose any condition and should not be used as a substitute for clinical assessment. Where session data suggests acute distress, the user is directed to appropriate support resources via the AI handoff close.

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Colophon

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Companion document: *V3KTOR Session File Format Specification v1.0*, Tobie David, 18 May 2026, deposited separately. Reference implementation: *v3ktor_eng.html* and *v3ktor_fr.html*, Tobie David, 18 May 2026.

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